

Element B

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Prior Attempts

- Failed 3D prints
- Structural failures
- Small details not getting shown

Our donor has brought up three critical concerns for our design and creation process. These three concerns are widely documented and equally important.

The first concern is the small details on the design won't be able to show once printed. This is disastrous for highly detailed chess pieces that require precision. Many consistently credible sources state that millimeter sized objects will break easily or won't be printed at all! (Atakok 2022).

The solution for this is to simply use larger measurements. Instead of creating small and precise pieces, we can sacrifice our precision for the structure of our pieces, allowing for stronger, more durable pieces (Syaefudin 2023).

Furthermore, on the subject of structure, many 3d printed pieces suffer from poor structural designs. This could mean that the printer struggles to print it, or that the structure is easily breakable. This will not be a concern for our project because we will design our prototypes with this

issue in mind, and ensure that our design is strong, compatible, and resilient (Lu, Lin, et al. 2014).

Finally, the 3d printer itself is an issue. There are many documented printing issues where the design isn't the problem, it's the printer. However our printer is quite reliable, and our sponsor is providing one that meets their preferences (Kuznetsov 2018).

Works Cited

- Atakok, Gurcan, et al. "Tensile, Three-Point Bending and Impact Strength of 3D Printed Parts Using PLA and Recycled PLA Filaments: A Statistical Investigation." *Journal of Materials Research and Technology*, vol. 18, May 2022, pp. 1542–1554, <https://doi.org/10.1016/j.jmrt.2022.03.013>.
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- Lu, Lin, et al. "Build-To-Last." *ACM Transactions on Graphics*, vol. 33, no. 4, 27 July 2014, pp. 1–10, <https://doi.org/10.1145/2601097.2601168>.
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